

1 Solution — GIAM 6.6

Problem 5

1.1 Problem

The helix in $\mathbb{R}^3$ is

$$H = \{(\cos t, \sin t, t) : t \in \mathbb{R}\}.$$

Find $\pi_{12}(H)$, $\pi_{13}(H)$, and $\pi_{23}(H)$, where $\pi_{ij} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ keeps the i -th and j -th coordinates and drops the third.

1.2 Setup

The three projections strip away one coordinate:

$$\pi_{12}(x, y, z) = (x, y), \quad \pi_{13}(x, y, z) = (x, z), \quad \pi_{23}(x, y, z) = (y, z).$$

For each $t \in \mathbb{R}$, the helix passes through $(\cos t, \sin t, t)$: the (x, y) -coordinates trace the unit circle while z rises linearly with t . The projections turn this 3D motion into three 2D “shadows” — one viewed from each coordinate-axis direction.

1.3 $\pi_{12}(H)$ — the Unit Circle

Apply π_{12} pointwise to H :

$$\pi_{12}((\cos t, \sin t, t)) = (\cos t, \sin t).$$

As t ranges over $\mathbb{R}$, the point $(\cos t, \sin t)$ traces the unit circle (each point hit infinitely often, once per rotation). So

$$\pi_{12}(H) = \{(\cos t, \sin t) : t \in \mathbb{R}\} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} = S^1.$$

This is the *shadow* of the helix looking straight down the z -axis — the helix’s xy -footprint is exactly its underlying circle of revolution.

1.4 $\pi_{13}(H)$ — the Curve $x = \cos z$

Apply π_{13} :

$$\pi_{13}((\cos t, \sin t, t)) = (\cos t, t).$$

Reading the second coordinate as z (since $z = t$),

$$\pi_{13}(H) = \{(\cos t, t) : t \in \mathbb{R}\} = \{(x, z) \in \mathbb{R}^2 : x = \cos z\}.$$

That is, the graph of $x = \cos z$ — a cosine curve “lying on its side,” with the height z on the vertical axis and the horizontal coordinate x oscillating between -1 and 1 .

This is the shadow viewed from the y -axis direction: the spiraling helix, seen edge-on, traces a sinusoid in the xz -plane.

1.5 $\pi_{23}(H)$ — the Curve $y = \sin z$

Apply π_{23} :

$$\pi_{23}((\cos t, \sin t, t)) = (\sin t, t).$$

Renaming again,

$$\pi_{23}(H) = \{(\sin t, t) : t \in \mathbb{R}\} = \{(y, z) \in \mathbb{R}^2 : y = \sin z\}.$$

The graph of $y = \sin z$ — sine curve on its side. The shadow viewed from the x -axis direction.

1.6 Summary

Projection	Set	Description
$\pi_{12}(H)$	$\{(\cos t, \sin t) : t \in \mathbb{R}\}$	unit circle S^1 : $x^2 + y^2 = 1$
$\pi_{13}(H)$	$\{(\cos t, t) : t \in \mathbb{R}\}$	curve $x = \cos z$ in the xz -plane
$\pi_{23}(H)$	$\{(\sin t, t) : t \in \mathbb{R}\}$	curve $y = \sin z$ in the yz -plane

Table 1.1

The three shadows together encode the full 3D structure of H : the circular projection captures the rotational sweep around the z -axis, and the two sinusoidal projections capture the linear ascent paired with that rotation.

1.7 Remark — Sine vs. Cosine, 90° Apart

The two side projections, $\pi_{13}(H)$ and $\pi_{23}(H)$, are *the same curve up to a $\pi/2$ shift in z* — because $\sin(t) = \cos(t - \pi/2)$.

Geometrically: looking at the helix from the y -axis, you see the x -coordinate of the rotating point as a function of height; looking from the x -axis, you see the y -coordinate, but the rotation is now "90° behind." The 90° rotation around the z -axis (between the two viewing directions) is exactly the $\pi/2$ phase shift between $\cos$ and $\sin$. Two views of the same motion, offset in time.